

Megha Srivastava

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I study how AI can support human learning and augment human capability. I approach this across domains, including programming education and code security, AI-assisted instruction for physical skills, and mathematical and scientific discovery. My work has been recognized by an ICML Best Paper Runner-Up Award, supported by the NSF and IBM PhD Fellowships, and appears in venues including Nature, NeurIPS, and ICML.

Education

Stanford University • Stanford, CA present
PhD in Computer Science, co-advised by Dorsa Sadigh & Dan Boneh
Supported by the NSF Graduate Research Fellowship (2018 – 2023) and IBM PhD Fellowship (2023-2024)

Massachusetts Institute of Technology • Cambridge, MA Fall 2023 – Winter 2024
Visiting PhD student at the Computer Science & Artificial Intelligence Laboratory, hosted by Jacob Andreas

Stanford University • Stanford, CA Spring 2019
BS+MS in Computer Science with Honors and Distinction, Minor in Creative Writing
Research advised by Percy Liang & Tatsunori Hashimoto
Ben Wegbreit Prize for Best Undergraduate Honors Thesis in Computer Science
Bing Overseas Studies Program at Oxford University (Logic & Computability) in Spring 2017

Experience

Research Intern – Genesis Molecular AI Summer 2026
San Mateo, CA
• Deep learning research at the intersection of generative AI and drug discovery.

Member of Technical Staff (Contractor) – Anthropic present
San Francisco, CA
• Support research projects and writing.

PhD Intern – Hudson River Trading Winter 2026
New York, NY
• Applied machine learning techniques towards building financial models with the Algorithm Development team.

AI Resident – Microsoft Research Fall 2019 – Fall 2020
Redmond, WA
• Quantifying effects of nondeterminism in neural network training (Mentors: Besmira Nushi, Eric Horvitz).

Research Intern – Google Research Summer 2019
Los Angeles, CA
• Improving robustness of image understanding models with the Research & Machine Intelligence team.

Research Intern – ETH Zürich Learning & Adaptive Systems Group Summer 2018
Zürich, Switzerland
• Active learning and adaptive hypothesis testing (Mentors: Hoda Heidari, Andreas Krause)
• Supported by the ETH Zürich Student Summer Research Fellowship.

Research Assistant – Vision & Perception Neuroscience Group Summer 2016
Stanford, CA
• Generalization and perceptual invariances in both human and artificial vision models (Mentor: Kalanit Grill-Spector)
• Supported by the Bio-X Undergraduate Research Award.

Publications (See Google Scholar for full list, * denotes equal contribution)

Erez Badash*, Dan Boneh*, Ilan Komargodski*, **Megha Srivastava***. “Hawkeye: Reproducing GPU-Level Non-Determinism”, Ninth Annual Conference on Machine Learning and Systems (MLSys), 2026
Robust ML-Security

Alexis Ross*, **Megha Srivastava***, Jeremiah Blanchard, Jacob Andreas. “Modeling Student Learning with 3.8 Million Program Traces”, International Conference on AI in Education (AIED), 2026
AI for Education-NLP

Publications Cont. (See Google Scholar for full list, * denotes equal contribution)

Jan Betley*, Niels Warncke*, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, **Megha Srivastava**, Nathan Labenz, & Owain Evans “*Training Large Language Models on Narrow Tasks Can Lead to Broad Misalignment*”
Nature, 2026
[Robust ML·NLP](#)

Megha Srivastava, Cédric Colas, Dorsa Sadigh, Jacob Andreas. “*Policy Learning with a Language Bottleneck*”
Transactions in Machine Learning (TMLR), 2026
[NLP·Embodied AI ·Human-AI Interaction](#)

Megha Srivastava*, Reihaneh Iranmanesh*, Yuchen Cui, Deepak Gopinath, Emily Sumner, Andrew Silva, Laporsha Dees, Guy Rosman, Dorsa Sadigh. “*Shared Autonomy for Proximal Teaching*”, ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025.
[AI for Education ·Embodied AI](#)

Megha Srivastava, Simran Arora, Dan Boneh. “*Optimistic Verifiable Training by Controlling Hardware Nondeterminism*”,
Advances in Neural Information Processing Systems 38 (NeurIPS) , 2024.
[Robust ML·Security](#)

Neil Perry*, **Megha Srivastava***, Deepak Kumar, Dan Boneh. “*Do Users Write More Insecure Code with AI Assistants?*” ACM Conference on Computer and Communications Security (CCS), 2023.
[mlsec.org Top-100 Computer Security Papers](#) 
[Security ·Human-AI Interaction](#)

Mina Lee, **Megha Srivastava**, Amelia Hardy, John Thickstun, Esin Durmus, Ashwin Paranjape, Ines Gerard-Ursin, Xiang Lisa Li, Faisal Ladhak, Frieda Rong, Rose E. Wang, Minae Kwon, Joon Sung Park, Hancheng Cao, Tony Lee, Rishi Bommasani, Michael Bernstein, Percy Liang “*Evaluating Human-Language Model Interaction*” Transactions of Machine Learning Research (TMLR), 2023.

Led experimental design; follow-up analysis and dataset release with John Thickstun 

§ **Question Answering**: Megha Srivastava, John Thickstun, Rose Wang, Minae Kwon, Mina Lee

§ **Crossword Puzzles**: Megha Srivastava

[Human-AI Interaction ·NLP](#)

Megha Srivastava, Noah Goodman, Dorsa Sadigh. “*Generating Language Corrections for Teaching Physical Control Tasks*”
Proceedings of the 40th International Conference of Machine Learning (ICML) , 2023.
[AI for Education ·Embodied AI](#)

Megha Srivastava, Erdem Biyik, Suvir Mirchandani, Noah Goodman, Dorsa Sadigh. “*Assistive Teaching of Motor Control Tasks to Humans*” Advances in Neural Information Processing Systems 36 (NeurIPS) , 2022.
[AI for Education ·Embodied AI](#)

Siddharth Karamcheti*, **Megha Srivastava*** Percy Liang, Dorsa Sadigh. “*LILA: Language-Informed Latent Actions*”
Conference on Robot Learning (CoRL), 2021.
[Embodied AI·NLP](#)

Megha Srivastava and Noah Goodman. “*Question Generation for Adaptive Education*” Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), 2021.
[AI for Education·NLP](#)

Megha Srivastava, Tatsunori Hashimoto, Percy Liang. “*Robustness to Spurious Correlations via Human Annotations*”
Proceedings of the 37th International Conference on Machine Learning (ICML), 2020.
[Robust ML·NLP·Human-AI Interaction](#)

Megha Srivastava, Besmira Nushi, Ece Kamar, Shital Shah, Eric Horvitz “*An Empirical Analysis of Backward Compatibility in Machine Learning Systems*” Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2020.
[Robust ML](#)

Megha Srivastava, Hoda Heidari, Andreas Krause. “*Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning.*” Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2019.
[Human-AI Interaction](#)

Tatsunori Hashimoto, **Megha Srivastava**, Hongseok Namkoong, Percy Liang. “*Fairness Without Demographics in Repeated Loss Minimization.*” Proceedings of the 35th International Conference on Machine Learning (ICML), 2018.
[Best Paper Runner-Up Award](#)
[Robust-ML](#)

Other Works (Blog Posts, Abstracts, Technical Reports, etc.)

- **Megha Srivastava** and Claude Sonnet 3.5. "Echo: A multi-agent AI system for patient-centered pharmacovigilance." 1st Open Conference of AI Agents for Science, 2025. **Spotlight Talk**. [Experimental conference exploring human-AI collaboration in research. More details in blog post on research process [🔗](#)]
- Sam Bowman, **Megha Srivastava**, Jon Kutasov, Rowan Wang, Trenton Bricken, Benjamin Wright, Ethan Perez, and Nicholas Carlini. "Findings from a Pilot Anthropic–OpenAI Alignment Evaluation Exercise." Anthropic, 2025. [🔗](#)
- Rose Wang and **Megha Srivastava**. "Productive Struggle: The Future of Human Learning in the Age of AI." The Stanford AI Lab Blog, 2025. Featured and translated in Spanish by the Inter-American Development Bank. [🔗](#)
- **Megha Srivastava** and John Thickstun. "Observations from HALIE: A Closer Look at Human-LM Interactions in Information-Seeking Contexts." Center for Research on Foundation Models Blog, 2023. [🔗](#)
- Kristin Lauter*, Cathy Yuanchen Li*, Krystal Maughan*, Rachel Newton*, **Megha Srivastava***. "Machine learning for modular multiplication." Proceedings of the WIN6 Workshop at Banff International Research Station for Mathematical Innovation and Discovery, 2023.
- **Megha Srivastava**, David Remus, Kalanit Grill-Spector. "The Role of Learning in Complex Object Recognition and Discrimination Across Spatial Transformations: An Experimental Comparison of Artificial CNNs and Human Subjects." Vision Sciences Society (VSS), 2017.

Selected Awards & Fellowships

- Human Robot Interaction (HRI) Pioneers, 2025
- Rising Star in Machine Learning (University of Maryland), 2023
- IBM PhD Fellowship, 2023
- American Association for the Advancement of Science (AAAS) Mass Media Fellowship (declined), 2019
- International Conference on Machine Learning (ICML) Best Paper Runner-Up Award, 2018
- National Science Foundation Graduate Research Fellowship, 2018
- Ben Wegbreit Prize for Best Undergraduate Honors Thesis in Computer Science (Stanford), 2018
- Phi Beta Kappa, 2018
- Tau Beta Pi, 2018
- Bio-X Research Award (Stanford), 2016
- Lunsford Award for Oral Presentation of Research Finalist (Stanford), 2016

Service

- Science Small Groups Mentor (2024)
- Stanford AI Lab Blog Editor, 2020-present
- Student Program Chair for Women in Machine Learning (WiML) 2023, co-located with NeurIPS 2023
- Reviewer for NeurIPS (2020 - curr.), ICLR (2021 - curr.), ICML (2021 - curr.), RA-L 2022, CoRL 2023, L4DC 2024, CHI 2025, HRI-LBR 2026

Teaching and Mentoring

- Course Assistant for CS 224R (Deep Reinforcement Learning, 2026)
- Course Assistant for CS 255 (Introduction to Cryptography, 2025)
- Course Assistant for CS 221 (Introduction to Artificial Intelligence, 2018)
- Instructor for Stanford Splash (Demystifying Hot Topics in Computer Science , 2017)
- Mentored research of undergraduate and MS students: *Reihaneh Iranmanesh* (Amherst College → PhD at Georgetown University, co-author on HRI'25 paper), *Zhiyin Lin* (Stanford → OpenAI, CURIS program), *Mallika Parulekar* (Stanford), *Jonathan Ouyang* (UCLA), *Benita Wong* (Stanford)

Guest Lectures

1. "Security of Modern Machine Learning Systems" AI4All, June 2025
2. Beyond Instruction Following: Language and Human-Robot Interaction *Stanford University* (Guest Lecture for CS 329X: Human-Centered NLP), October 2024
3. "Do Users Write More Insecure Code with AI Assistants?" *University of Maryland College Park* (Guest Lecture for Large Language Models, Security, and Privacy seminar), October 2023

1. "Modeling Student Learning with 3.8 Million Program Traces" *Together AI*, February 2026
2. "Modeling Student Learning with 3.8 Million Program Traces" *Deep Learning: Classics and Trends (ML Collective)*, November 2025
3. "Security of Modern Machine Learning" *Accenture Responsible AI Course*, September 2025
4. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *Google Privacy in ML Seminar*, February 2025
5. "Security of Modern Machine Learning" *DaVita x Stanford Human-Centered AI*, February 2025
6. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *AI and Cryptography Workshop, Joint Mathematics Meeting*, January 2025
7. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *University of Washington Security Seminar*, January 2025
8. "New Challenges of Trust with Large-Scale AI Systems" *University of Chicago*, November 2024.
9. "Security of Modern Machine Learning" *Stanford Cybersecurity & Privacy Festival*, October 2024
10. "Policy Learning with a Language Bottleneck" **Spotlight Talk** at *Training Agents with Foundation Models Workshop (RLC 2024)*, August 2024.
11. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *University of Toronto*, July 2024
12. "Security of Modern Machine Learning" *Accenture Responsible AI Course*, July 2024
13. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *Stanford Security Forum*, April 2024
14. "Challenges in Human-AI Interaction for Information-Seeking Tasks" *Rising Stars in Machine Learning Workshop*, November 2023
15. "Robustness to Spurious Correlations via Human Annotations" *Two Sigma PhD Symposium*, June 2023.
16. "Do Users Write More Insecure Code with AI Assistants?" *UC Berkeley Security Seminar*, May 2023
17. "Assistive Teaching of Motor Control Tasks to Humans" *Stanford Collaborative Haptics in Robotics and Medicine Lab*, May 2023.
18. "Assistive Teaching of Motor Control Tasks to Humans" *Simons Institute Workshop on AI & Humanity*, July 2022.
19. "Assistive Teaching of Motor Control Tasks to Humans" *SystemX Alliance Fall Conference*, November 2022.
20. "Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning" *Oxford University Algorithms at Work Reading Group*, May 2021.
21. "Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning" *Microsoft Research AI & Society Reading Group*, October 2019.
22. "Fairness & Robustness with Missing Information" *Stanford Causality & Cognition Lab*, December 2020.